



ganit

data speaks

PHARMA | GCP + REACT/NODE

Sun Pharma – GCP Data Engineering & Frontend Analytics Platform

Scalable GCP-native Data Pipelines + React/Node.js Self-Serve Analytics for Global Pharma Operations

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Ganit Confidential

Sun Pharma GCP Data Eng & Frontend Analytics

3x

Faster Pipeline Throughput

20+

GCP Services Integrated

70%

Infra Cost Reduction

1000+Concurrent Dashboard
Users

Unifying Pharma Manufacturing, Commercial & Regulatory Data on GCP with a React/Node.js Self-Serve Analytics Portal

Problem Statement	Solution Overview	Business Value
▶ Siloed SAP, Veeva & LIMS data on legacy systems ▶ No real-time manufacturing analytics on cloud ▶ Static reports with no self-serve capability ▶ Fragmented frontend BI tools per department	▶ GCP-native lakehouse (BigQuery + Dataflow) ▶ Cloud Composer DAG orchestration ▶ React.js + Node.js analytics portal ▶ Looker Studio + custom dashboards	▶ Real-time batch release & QC dashboards ▶ 3x pipeline throughput improvement ▶ Self-serve analytics for 1000+ global users ▶ 70% infra cost reduction vs on-prem

GCP Services: BigQuery | Dataflow | Cloud Composer | Pub/Sub | GCS | Dataproc | Vertex AI | Cloud Run | Looker

Frontend: React.js | Node.js | Express | TypeScript | D3.js | Recharts | REST APIs | JWT Auth

Outcome

Data Engineering & Pipeline Speed

GCP Dataflow + Cloud Composer pipelines replaced legacy Informatica jobs — achieving 3x pipeline throughput and reducing batch processing windows from 8 hours to under 90 minutes.

Cost Optimisation

Decommissioned 5 on-premise Hadoop clusters by migrating to BigQuery + GCS — achieving 70% reduction in infrastructure costs with auto-scaling for quarter-end peaks.

Frontend & User Experience

Built a React.js + Node.js analytics portal with role-based access, 15 custom dashboards, and D3.js visualisations — enabling 1000+ concurrent users with sub-2s page load times.

Regulatory & Compliance

Automated Schedule H, pharmacovigilance, and FDA 21 CFR Part 11 audit trail generation through GCP Dataform + BigQuery — cutting compliance report turnaround from T+5 to same-day.

Problem Statement

Sun Pharma operates across 40+ markets with a complex stack of SAP S/4HANA, Veeva Vault, LIMS (LabWare), MES (Syncade), and third-party market data feeds. Data resided in on-premise Hadoop clusters with no cloud strategy, and analytics relied on static Excel/SSRS reports — with no modern frontend for self-serve insights.

Challenge 1

01

Legacy On-Prem Hadoop Infrastructure

5 on-premise Hadoop clusters running Informatica ETL jobs — causing 8-hour batch windows, frequent job failures, and no elasticity for peak processing loads.

Challenge 2

02

Fragmented Data Sources & No Cloud Ingestion

SAP S/4HANA, Veeva Vault, LIMS, and MES data were siloed with no unified GCP ingestion layer — causing inconsistent KPIs across commercial and manufacturing teams.

Challenge 3

03

No Modern Frontend Analytics Portal

Analytics were delivered via static SSRS/Excel reports emailed to teams — no interactive dashboards, no self-serve filtering, and no mobile-responsive access for field teams.

Challenge 4

04

Compliance & Audit Trail Gaps

FDA 21 CFR Part 11, Schedule H, and pharmacovigilance reports required manual data collation from 6+ systems — creating T+5 delays and high risk of submission errors.

Solution Overview

GCP Data Engineering Architecture

SAP S/4HANA & Veeva Vault ingested via Cloud Dataflow with CDC using Pub/Sub event streaming.

LIMS & MES data processed via Cloud Composer (Airflow) DAGs with schema validation and SLA alerting.

GCS Bronze → Silver → Gold lakehouse zones; Dataproc Spark for heavy transformations.

BigQuery DWH as serving layer; Dataform for SQL transformations and dbt-style lineage.

Frontend Portal — React.js + Node.js

- ▶ React.js SPA with TypeScript + Recharts/D3.js visualisations
- ▶ Node.js/Express REST API layer with JWT auth & RBAC
- ▶ 15 interactive dashboards; <2s load time; mobile-responsive

Source Systems

SAP S/4HANA | Veeva Vault | LIMS (LabWare) | MES (Syncade) | Market Data Feeds

CLOUD PUB/SUB + CLOUD DATAFLOW (Stream & Batch Ingestion)

CLOUD COMPOSER (Airflow DAGs) + DATAPROC SPARK (ETL & Transforms)

GCS Lakehouse — Bronze → Silver → Gold

BIGQUERY DWH + DATAFORM (SQL Transforms + Lineage)

REACT.JS + NODE.JS ANALYTICS PORTAL — Commercial | Manufacturing | Regulatory | QC | Doctor 360

12+ TB Data Migrated | 40+ Markets | 1000+ Concurrent Users | 20 GCP Services | 150+ Pipelines

GCP Data Engineering Domain**Ingestion & Streaming**

- Cloud Pub/Sub for SAP change events
- Dataflow Apache Beam streaming jobs
- Dead-letter queue + alerting via Cloud Monitoring
- Schema registry with Protobuf/Avro validation

Batch ETL & Transformation

- Cloud Composer Airflow DAGs (150+ pipelines)
- Dataproc Spark for heavy pharma data joins
- Dataform SQL models with unit test framework
- Incremental & full-refresh load strategies

GCS Lakehouse Architecture

- Bronze/Silver/Gold zone partitioning on GCS
- Delta-equivalent using BigQuery time-travel
- Dataplex for zone governance & data quality
- Auto-schema evolution with policy tags

BigQuery Analytics DWH

- Star schema models for Commercial & Supply Chain
- Materialised views for sub-second dashboard queries
- Slot reservations for guaranteed QoS at peak load
- Row-level security by market & business unit

ML & Vertex AI

- Demand forecasting models on Vertex AI Workbench
- Feature Store for reusable pharma ML features
- AutoML for batch QC anomaly detection
- Model monitoring & drift detection pipelines

Governance & Observability

- Dataplex automated data classification & tagging
- Data Catalog with 600+ business attribute entries
- Cloud Monitoring dashboards + PagerDuty alerts
- End-to-end lineage from SAP to BigQuery table

Frontend Analytics Domain

Frontend Architecture & Engineering Pillars

React.js SPA (TypeScript)

Component-driven architecture with React 18, TypeScript, Context API + React Query for server state; lazy-loaded dashboard modules reducing initial bundle size by 60%.

Node.js / Express API Layer

RESTful API server with Express.js; middleware for JWT authentication, RBAC enforcement, rate limiting, and BigQuery read optimization via connection pooling.

Data Visualisation & UX

D3.js for geospatial maps, Recharts for time-series/bar charts, AG Grid for paginated data tables; WCAG 2.1 AA accessible components; mobile-first responsive layouts.

Security & Deployment

Google Cloud Run containerised deployment with Cloud Armor WAF; OAuth 2.0 + Google Identity for SSO; Cloud CDN for static assets; 99.95% SLA uptime.

15 Dashboard Modules Delivered

Commercial Sales & Market Share

Commercial

Doctor 360 & Prescription Trends

CRM

Territory & Rep Performance

Field Force

Manufacturing Batch Release Tracker

Mfg

QC / QA Deviation & CAPA Monitor

Quality

Supply Chain & Inventory Health

SCM

Demand Forecasting & S&OP

Planning

Regulatory Submission Tracker

Compliance

+ 7 more modules: Financial P&L, API Mfg, Clinical, Market Access, HR, ESG & Sustainability

⚡ BigQuery cold query latency for large datasets

✓ Materialised views + partitioned tables cut P95 load time from 8s to 1.4s.

⚡ Multi-tenant data isolation for global markets

✓ Row-level security via Node.js middleware + BigQuery column-level policy tags per market/BU.

Ganit's Deliverables and the Value Generated for Sun Pharma

Deliverables & Value Generated

Deliverables

GCP Platform Setup & Migration: Provisioned GCP landing zone with VPC, IAM, Dataplex zones, and migrated 12 TB of historical data from 5 on-premise Hadoop clusters to GCS Bronze layer.

Data Pipeline Engineering: Built 150+ Cloud Composer DAGs and Dataflow jobs ingesting SAP S/4HANA, Veeva Vault, LIMS, MES, and market data — with DQ rules, retry logic, and SLA monitoring.

BigQuery Data Warehouse: Designed star schema models with Dataform SQL transforms, materialised views, row-level security, and slot reservations for guaranteed performance at quarter-end peaks.

React/Node.js Analytics Portal: Delivered a 15-module SPA with JWT auth, RBAC, D3.js/Recharts visualisations, REST API layer (Node/Express), Cloud Run deployment, and CDN-optimised static assets.

Governance & Observability: Implemented Dataplex data catalog with 600+ attributes, Cloud Monitoring dashboards, PagerDuty alerting, end-to-end lineage, and FDA 21 CFR Part 11 audit trail automation.

Value Generated

3x Faster

Pipeline Throughput

Replaced 8-hour Informatica batch jobs with GCP Dataflow streaming — delivering data to BigQuery in under 90 minutes end-to-end.

70% Lower

Infrastructure Cost

Decommissioned 5 Hadoop clusters — migrating to serverless BigQuery + GCS with auto-scaling for cost-efficient peak processing.

<2s Load

Dashboard Response

Materialised views + partitioned BigQuery tables + Cloud CDN delivered sub-2-second P95 load times across all 15 dashboard modules.

1000+

Concurrent Portal Users

Cloud Run autoscaling + Node.js connection pooling supports 1000+ simultaneous users with zero performance degradation.

Same-Day

Compliance Reporting

FDA 21 CFR Part 11, Schedule H, and pharmacovigilance reports automated end-to-end — replacing T+5 manual collation processes.